

# CS 486/686

# Markov Decision Processes I

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Lecture 13

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Learning

## Learning goals

- Motivate modeling a decision problem as a **Markov decision process**.
- Describe the **components** of a fully-observable MDP.
- Explain why we use a **discounted** reward function.
- Define the **policy** of an MDP.
- Give examples of how the **reward function** shapes the optimal policy.

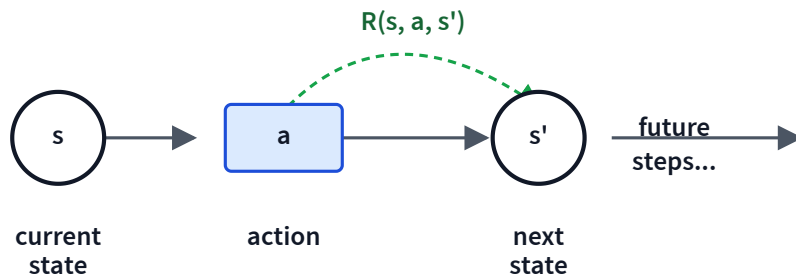
# From one-shot decisions to ongoing ones

## Two new things

- **Horizon:** the problem may never end.
  - *Infinite horizon:* the process may go on forever.
  - *Indefinite horizon:* the agent will stop eventually, but not at a known time.
- **Rewards along the way**, not just utility at the end.
  - The agent may never reach an "end".
  - Each step's reward includes action costs + bonuses/penalties.

A Markov decision process (MDP) formalizes this.

# A Markov decision process



An MDP is a 4-tuple  $(S, A, P, R)$ :

- $S$ : set of states.
- $A$ : set of actions.
- $P(s' \mid s, a)$ : transition probabilities.  
*Stationary*: same for every  $t$ .
- $R(s, a, s')$ : reward function.

For this lecture we'll use the simpler form  $R(s)$  = reward for entering state  $s$ .

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# How do we aggregate rewards over time?

## Total reward

$$\sum_{t=0}^{\infty} R(S_t)$$

May diverge  $\Rightarrow$   
policies are not  
comparable.

## Average reward

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=0}^n R(S_t)$$

Zero whenever the total  
reward is finite.

## Discounted reward

$$\sum_{t=0}^{\infty} \gamma^t R(S_t), \quad \gamma \in [0, 1)$$

Always finite; the standard  
choice.

The discounted sum is also called the **return**  $G(S_0)$ .

# The discount factor $\gamma$

- **Time preference.** A dollar today is worth more than a dollar tomorrow.

- **Finiteness.** If  $R(s) \in [-R_{\max}, R_{\max}]$  and  $\gamma \in [0, 1)$ :

$$|\sum_{t=0}^{\infty} \gamma^t R(S_t)| \leq \frac{R_{\max}}{1 - \gamma}.$$

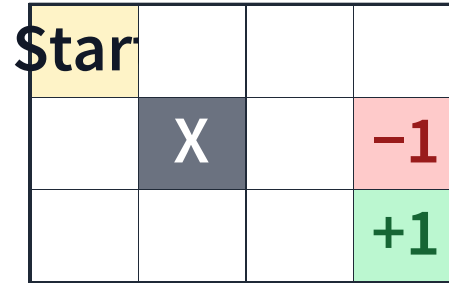
- **Effective horizon.** Roughly  $\frac{1}{1-\gamma}$  steps matter (e.g.  $\gamma = 0.9 \Rightarrow$  about 10 steps).

## Variants

- **Fully-observable MDP** — the agent always knows its current state. (Today.)
- **Partially observable MDP (POMDP)** — combine MDP + HMM; the state is hidden behind observations.

# A $3 \times 4$ grid world

What should the robot do to maximize its rewards?



- Let  $s_{ij}$  be row  $i$ , column  $j$ .  $s_{11}$  is the start.
- $s_{22}$  is a wall.  $s_{24}$  and  $s_{34}$  are goal states — the robot exits the world there.

# The grid world as an MDP

|      |   |  |    |
|------|---|--|----|
| Star |   |  |    |
|      | X |  | -1 |
|      |   |  | +1 |

**Actions:** up / down / left / right (in every state).

**Transition**  $P(s' \mid s, a)$ :

- 0.8 — action takes intended effect.
- 0.1 — 90° left of intended.
- 0.1 — 90° right of intended.
- If the robot bumps a wall, it stays put.

**Reward**  $R(s)$ :  $R(s_{24}) = -1$ ,  $R(s_{34}) = +1$ ; otherwise  $R(s) = -0.04$ .

# Understanding the transition model

**Q1.** The robot is in  $s_{14}$  (top-right corner) and tries to move right. What is the probability it *stays* in  $s_{14}$ ?

|      |   |  |          |
|------|---|--|----------|
| Star |   |  | $s_{14}$ |
|      | X |  | -1       |
|      |   |  | +1       |

- A. 0.1
- B. 0.2
- C. 0.8
- D. 0.9
- E. 1.0

**D — 0.9.** Trying right: 0.8 it bumps the wall (stays) + 0.1 it veers up (also bumps wall, stays) = 0.9. The remaining 0.1 veers down to  $s_{24}$ .

# Why a fixed action sequence is not enough

**Q2.** In a *deterministic* env, the sequence *down, down, right, right, right* is optimal for the grid world.

**True.** Five steps take us from  $s_{11}$  to  $s_{34}$ .

**Q3.** In the *stochastic* env, the same sequence may end at *more than one* square.

**True.** E.g. five right-veers in a row send us to  $s_{12}$ .

**Q4.** So in the stochastic env, a fixed sequence is still good enough.

**False.** Actions misfire — we need to *re-plan* from whatever state we end up in.

|      |   |  |    |
|------|---|--|----|
| Star |   |  |    |
|      | X |  | -1 |
|      |   |  | +1 |

Need an action for **every state** — a **policy**.

# A policy

## Policy

A **policy**  $\pi$  specifies the agent's action as a function of the current state.

- **Stationary** — depends only on the state:  $\pi : \mathcal{S} \rightarrow \mathcal{A}$ .
- **Non-stationary** — also depends on the time step:  $\pi : \mathcal{S} \times \mathbb{N} \rightarrow \mathcal{A}$ .

For an infinite-horizon fully-observable MDP, the optimal policy is **stationary**.

# The reward function reshapes the optimal policy

Same dynamics; different  $R(s)$  for non-goal states — very different optimal behaviors.

We will look at five qualitatively different reward regions:

1.  $R(s) < -1.6284$  — *life is painful*
2.  $-0.4278 < R(s) < -0.0850$  — *life is quite unpleasant*
3.  $R(s) = -0.04$  — *life is unpleasant* (the default)
4.  $-0.0221 < R(s) \leq 0$  — *life is only slightly dreary*
5.  $R(s) > 0$  — *life is GOOD*

Each region balances **risk** (falling into  $-1$ ) against **reward** (reaching  $+1$  quickly).

# Five reward regions, five optimal policies

painful

$$R < -1.6284$$

|   |   |   |    |
|---|---|---|----|
| → | → | → | ↓  |
| ↓ | X | → | -1 |
| → | → | → | +1 |

head to nearest  
exit, even -1

quite unpleasant

$$-0.43 < R < -0.085$$

|   |   |   |    |
|---|---|---|----|
| ↓ | → | ↓ | ←  |
| ↓ | X | ↓ | -1 |
| → | → | → | +1 |

shortcut through  
 $s_{13}$

unpleasant

$$R = -0.04$$

|   |   |   |    |
|---|---|---|----|
| ↓ | ← | ← | ←  |
| ↓ | X | ↓ | -1 |
| → | → | → | +1 |

long way around;  
avoid -1

slightly dreary

$$-0.022 < R \leq 0$$

|   |   |   |    |
|---|---|---|----|
| ↓ | ← | ← | ↑  |
| ↓ | X | ← | -1 |
| → | → | → | +1 |

no risk; bang  
against walls

GOOD

$$R > 0$$

|   |   |   |    |
|---|---|---|----|
| ⊕ | ⊕ | ⊕ | ↑  |
| ⊕ | X | ← | -1 |
| ⊕ | ⊕ | ← | +1 |

wander forever;  
avoid both goals

Watch the action at  $s_{13}$  (top, third column): it changes with every region.

# The value function

## Expected utility of a policy

- $V^\pi(s)$ : expected utility of entering state  $s$  and following  $\pi$  thereafter.
- $V^*(s)$ : expected utility of entering  $s$  and following the **optimal** policy  $\pi^*$ .

|   |     |     |     |      |
|---|-----|-----|-----|------|
| 0 | .70 | .60 | .60 | .388 |
| 0 | .76 | X0  | .66 | -1   |
| 0 | .80 | .80 | .91 | +1   |

$V^*(s)$  for  $\gamma = 1, R(s) = -0.04$  on non-goal states.

# The Q-function and the optimal policy

## Q-function

$$Q^*(s, a) = R(s) + \sum_{s'} P(s' \mid s, a) V^*(s')$$

"If I were in  $s$  and took action  $a$  now, then followed  $\pi^*$  forever after — how good is my future?"

$$\pi^*(s) = \arg \max_a Q^*(s, a)$$

Given  $V^*$ , the optimal policy is one greedy step away.

# Optimal action at $s_{13}$ ?

**Q5.** Using  $Q^*(s, a) = R(s) + \sum_{s'} P(s' | s, a) V^*(s')$ , what is  $\pi^*(s_{13})$ ?

|   |     |     |     |      |
|---|-----|-----|-----|------|
| 0 | .70 | .60 | .60 | .388 |
| 0 | .76 | X   | .66 | -1   |
| 0 | .80 | .80 | .91 | +1   |

$s_{13}$  is the yellow cell ( $V^* = 0.611$ ).

Computing each Q-value (4 actions):

Left:  $-0.04 + 0.8 \cdot 0.655 + 0.1 \cdot 0.611 + 0.1 \cdot 0.660 = 0.6111$   
 Right:  $-0.04 + 0.8 \cdot 0.388 + 0.1 \cdot 0.611 + 0.1 \cdot 0.660 = 0.3975$   
 Down:  $-0.04 + 0.8 \cdot 0.660 + 0.1 \cdot 0.388 + 0.1 \cdot 0.655 = 0.5923$   
 Up:  $-0.04 + 0.8 \cdot 0.611 + 0.1 \cdot 0.655 + 0.1 \cdot 0.388 = 0.5531$

**C — Left.** Left has the highest  $Q^*$  (0.6111). It moves toward the safer states; the -1 to the right is too close.

## Learning goals (recap)

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- ✓ Explain why we use a **discounted** reward function.
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**Next: how do we compute  $V^*$ ?**

Today: given  $V^*$ , the optimal policy is just an argmax. But how do we *find*  $V^*$  in the first place?

**L14: Bellman equations & value iteration.**